



Assignment 2

Classifying Digits

Objective: The goal of this assignment is to understand the effectiveness of gradient descent in training a simple neural network and compare its performance with logistic regression on a real dataset, using the starter code provided in section 4 materials(Assignment2.ipynb).

Problem Statement: You are given a dataset of handwritten digits (MNIST-like, using the `digits` dataset from `sklearn`). Your task is to build and train a simple neural network using gradient descent to classify the digits. Additionally, you will train a logistic regression model on the same dataset and compare the accuracy of both models, **do not use any built-in libraries to build your Neural networks (no Pytorch/TensorFlow,.....)**.

Steps to Complete the Assignment:

Step 1: Load and Preprocess the Data

1. Import the `digits` dataset from `sklearn.datasets`.
2. Normalize the input features using `StandardScaler` to ensure effective learning.
3. Convert the target labels into one-hot encoding to be used in the neural network.
4. Split the dataset into training and testing sets.

Step 2: Implement a Simple Neural Network

1. Define a `NeuralLayer` class with the following methods:
 - `__init__()`: Initializes the weight matrix with small random values.
 - `sigmoid()`: Implements the sigmoid activation function and its derivative.
 - `forward()`: Computes the output of the layer.
 - `backward()`: Updates the weights using gradient descent.
2. Train the neural network for a specified number of epochs (e.g., 1000 epochs).
3. Evaluate the neural network by predicting the test set labels and computing accuracy.

Step 3: Train and Evaluate Logistic Regression

1. Train a logistic regression model using `LogisticRegression` from `sklearn`.
2. Use the training data to fit the logistic regression model.



3. Predict the labels for the test dataset and compute accuracy.

Step 4: Compare the Results

1. Print the accuracy of both models.
2. Discuss which model performs better and why.
3. Reflect on how gradient descent optimizes the neural network's performance compared to logistic regression.

Bonus: Try adjusting the learning rate or the number of epochs and observe how it affects the accuracy.

Submission Requirements:

- Submit a Python notebook (.ipynb) implementing all the above steps.
- Include comments explaining each part of your code.
- Provide a short report (1-2 paragraphs) summarizing your observations and comparison results inside your **notebook** (use Markdown).